Accessing & Using the Data

Step-by-step guide to obtaining and loading the repository data

All data is committed directly into the repository — there is no external download or database. You obtain it by cloning, and use it by reading the files with pandas (and geopandas for the maps).

1 Install Git LFS, then clone

Three large files are stored via Git LFS and must be fetched with LFS installed first:

- `DATA/raw/ml_dataset_monthly.parquet` (~35 MB)
- `DATA/shapefiles/gadm/africa_adm2_combined.gpkg` (~62 MB)
- `DATA/shapefiles/gadm/africa_adm0_basemap.gpkg` (~37 MB)

```
git lfs install
git clone https://github.com/News-Signal-for-Food-Insecurity/news-signal-africa.git
cd news-signal-africa
```

TIP
Already cloned without LFS? Fetch the real files in place with `git lfs pull`.

2 Know the file formats

- **DATA/dataset.parquet** — the main modelling table (Parquet, pandas).
- **DATA/raw/*.parquet, DATA/modelling/*.parquet** — source & feature tables (Parquet).
- **DATA/filtering/*.csv, results/**/*.csv** — CSV.
- **DATA/*.json, results/**/*.json** — JSON metadata & metrics.
- **DATA/shapefiles/gadm/*.gpkg** — GeoPackage (geopandas).

3 Load the main dataset

DATA/dataset.parquet is the table every training script reads: 3,245 rows covering 353 districts across 18 countries, Oct 2020 – Feb 2024. Each row is one district-period observation.

```
import pandas as pd

df = pd.read_parquet("DATA/dataset.parquet")
print(df.shape)             # (3245, ...)
print(df.columns.tolist())
print(df.head())
```

Key columns

- **Target:** `target_crisis_binary` — IPC Phase ≥ 3 crisis 8 months ahead (about 28% positive).
- **AR features (5):** ipc_lag_1, ipc_persistence_2yr, spatial_lag, ipc_period, ipc_country.
- **News features (18):** {theme}_relative_coverage and {theme}_zscore for 9 themes (conflict, displacement, economic, food_security, governance, health, humanitarian, weather, other).

Leakage caveat — read before reusing the z-scores

The `{theme}_zscore` columns in dataset.parquet are **fold-aware**: they were standardised within each training window for the paper's CV split. If you build your own model with a different split, do **not** reuse these z-scores directly — recompute them from the raw monthly counts to avoid leakage:

```
import pandas as pd

monthly = pd.read_parquet("DATA/modelling/monthly_gdelt_features.parquet")
# standardise within your own training folds only
```

4 Load the map layers (optional)

Join the GADM boundaries to dataset.parquet on the district/GADM identifier to build choropleths.

```
import geopandas as gpd

districts = gpd.read_file("DATA/shapefiles/gadm/africa_adm2_combined.gpkg")
countries = gpd.read_file("DATA/shapefiles/gadm/africa_adm0_basemap.gpkg")
```

5 Use the pre-computed results

results/ holds the committed model outputs as plain CSV/JSON — fold metrics, predictions, feature importance, and the null distribution — so you can analyse the findings without retraining anything.

```
import pandas as pd, json

folds      = pd.read_csv("results/window_2yr/fold_results.csv")
importance = pd.read_csv("results/window_2yr/feature_importance.csv")
summary    = json.load(open("results/window_2yr/metrics_summary.json"))
```

Reuse & licensing

Code is released under the **MIT License** (see the LICENSE file). The data is subject to **FEWSNET and GDELT terms of use**: you may freely read and analyse it, but if you redistribute it you must honour those upstream terms and cite the sources — FEWSNET IPC, GDELT GKG, and GADM v4.1.